Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/03/2006
PI Name:	Ronney, Paul D. Sc.D.		
Project Title:	Radiative Enhancement Effects on Flame Spread (REEFS): A Space Flight Experiment Proposal		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	90089-1453	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC3-1056		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
	The research is divided into three topics: 1) radiation-dominated flame spread over thick fuel beds, 2) concurrent-flow flame spread rates and extinction conditions effects of convective and radiative losses, and 3) addition of gaseous fuels to the oxidizing atmosphere. Flame spread over flat solid fuel beds is a useful means of understanding more complex, two-phase, non-premixed flames, such as those found in fires in enclosures, e.g., manned spacecraft and terrestrial buildings. Both transport and chemical effects are well known to affect the characteristics of flame spread. Conductive, convective and radiative transport processes may couple to each other and to chemical effects in unusual ways. Consequently, the goal of this work is to extend previous studies by employing μg experiments with emphasis on the atmospheres and flow environments likely to be present in fires which might occur at μg. The goal of this work is to provide information that will enhance crew safety in manned spacecraft including the Space Shuttle, International Space		

Task Description:	Station, and potential future missions to the moon and Mars. The understanding and control of accidental fires is a critical safety issue in both terrestrial and space-borne environments. Space environments are different from earth-based environments for several reasons. Of course, the major difference is the lack of buoyancy-induced convection. Another difference is that fires in spacecraft may occur in non-standard atmospheres because of the use of non-standard breathing atmospheres or the use of diluents as fire extinguishing agents. Consequently, the understanding of these effects is critical to understanding how fires might start, spread, and be extinguished at µg conditions. One of the tasks of this project, the study of CO2 as a fire extinguishing agent, is particularly relevant to fire safety on the International Space Station, whose fire suppression system uses CO2 as the extinguishing agent. Our experiments show that CO2 is less effective that other agents such as helium at microgravity, whereas the opposite is true at earth gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Radiation is the main mechanism of fire spread between buildings, thus understanding radiation effects on fires is useful to fire prevention and suppression on earth as well as in space.
Task Progress:	Flame spread tests were performed in the NASA-Glenn Zero Gravity Facility (ZGF) to define the test matrix for the REEFS space flight experiments in the FEANICS facility. Foam fuels were chosen for these tests to represent a thermally-thick solid fuel that could provide sufficiently high spread rates that meaningful data could be obtained within the µg time available in drop-tower experiments. Two types of thermally-thick, open-cell foams were tested: rigid polyphenolic (PP) foam and flexible polyurethane (PU) foam. Over 150 tests were conducted. The salient features of these results are: 1. At earth gravity all three diluents have nearly the same efficacy in reducing flame spread rates and causing flame extinction, for total pressures and oxygen mole fractions tested. 2. At microgravity, for higher total pressures and lower oxygen mole fractions, all three diluents have nearly the same efficacy in reducing flame spread rates and causing flame extinction 3. Most importantly, at microgravity, for lower total pressures and higher oxygen mole fractions, CO2 is clearly the least effective of the three diluents tested and He appears to be the most effective. 4. The spread rates in undiluted atmospheres is higher at microgravity, but dilution has a stronger effect on spread rates at µg than at 1g. At low pressures, at earth gravity, the spread rates are almost independent of pressure whereas at microgravity, the spread rates are substantially higher at higher pressure. This again shows the qualitative change in behavior between earth gravity and microgravity for thermally-thick fuels. At earth gravity, the efficacy of all three diluents for high O2 concentration / low total pressure conditions is similar, leading to speculation that again He may be the most effective diluent at µg. The effect of opposed flow velocity (U) on flame spread rate was measured. With 35% O2 in CO2 at 2 atm total pressure, the spread rate shows a slight minimum at an intermediate value of U (˜ 1 cm/sec). A minimum spread rate at intermediate U is consistent with the predictions of a theoretical model developed for this purpose. A series of tests were conducted to determine whether tests at earth gravity, for either upward or downward flame spread, are “conservative” in that the minimum oxygen mole fraction (LOI) supporting combustion is higher at microgravity than at earth gravity. If earth-gravity tests are “conservative,” then earth gravity flammability criteria may safely be used to screen materials to be used at microgravity conditions. Two different foam materials were used: polyphenolic green and polyurethane. A comparison of the LOI data for earth gravity (upward and downward flame spread) with microgravity conditions showed that of the cases tested only one, namely high-pressure O2-CO2 atmospheres, shows this non-conservative behavior. This is believed to be related to radiation effects, which are most prevalent at high pressure and with CO2 diluent. Most potentially flammable materials of interest, with the exception of paper, are thick enough that the material can be considered thermally-thick, that is, the thermal penetration depth into the fuel bed is smaller than the physical thickness of the fuel sample. The difficulty in assessing the flammability of thermally-thick materials at µg is that long durations of µg are needed since spread rates are slow and the corresponding characteristic time scales are long. REEFS skirts this problem by employing foam fuels that have much lower density and thermal conductivity than fuels conventionally used in thick-fuel flame spread experiments (e.g. PMMA), and thus much faster spread rates. In this way, some meaningful results can be obtained even in short-duration drop tower experiments. Even in space experiments where µg time is less of a consideration, dense fuel experiments present difficulties. Because of the large length and time scales, ignition energy requirements are large, and available test facilities such as CIR may have inadequate size test sections. Some space experiments may continuously feed a roll of thermally-thin fuel (e.g. paper) into the test section, with a feedback mechanism to maintain the flame front at a specified location. By using "compressible," flexible foam fuel such as REEFS' polyurethane foam, it is proposed that the roll-fed apparatus could be modified to incorporate thermally-thick foam fuels. Consequently, tests using polyurethane foam were conducted to determine if this flexible fuel could be deployed in the roll configuration. Results show that foam density changes less than 10% compared to the original values for practically all conditions tested.
Bibliography Type:	Description: (Last Updated: 01/29/2007)
Articles in Peer-reviewed Journals	Son Y, Zouein G, Ronney PD, Gokoglu S. "Comparison of Inert Fire Suppressants for Thick Solid Fuels at Microgravity." Submitted to Proceedings of the Combustion Institute, Vol. 31 (2006). , Dec-2005
Presentation	Son, Y.; Zouein, G.; Gokoglu, S.; Ronney, P. D. " Comparison of Carbon Dioxide and Helium as Fire Extinguishing Agents for Spacecraft" 4th Joint U.S. Sections Meeting, Combustion Institute, Philadelphia, PA Mar-2005